

Sometimes elements of the field $GF(2^m)$ are written in terms of a generator (i.e., a primitive root), g

$$g^{q-1} = g^0$$

- If g is a generator of $GF(2^m)$, then the field can be represented by $\{0, g^0, g^1, g^2, \dots, g^{q-3}, g^{q-2}\}$, where q is the order of the field
- Often, the generator is taken to be the element that satisfies $f(g) = 0$, where $f(x)$ is the irreducible polynomial that is chosen to form the field

E.g., consider $GF(2^3)$, as defined with the irreducible polynomial $x^3 + x + 1$

We want g to be the element for which $f(g) = 0$

$$g^3 + g + 1 = 0$$

$$\Rightarrow g^3 = -g - 1 \quad (\text{in } 2^m)$$

$$g^0 = 1 \quad 000$$

$$g^1 = g \quad 001$$

$$g^2 = g^2 \quad 010$$

$$g^3 = g + 1 \quad 011$$

$$g^4 = g(g^3) = g(g + 1) = g^2 + g \quad 110$$

$$g^5 = g(g^4) = g(g^2 + g) = g^3 + g^2 = (g + 1) + g^2 = g^2 + g + 1 \quad 111$$

$$g^6 = g(g^5) = g(g^2 + g + 1) = g^3 + g^2 + g = (g + 1) + g^2 + g = g^2 + 1 \quad 101$$

$$g^7 = g(g^6) = g(g^2 + 1) = g^3 + g = (g + 1) + g = 1$$

$$g + g = 2g = 0 \pmod{2}$$

Using this representation,

- Addition of elements: Simple XOR

$$g^3 + g^5 = 011 \oplus 111 = 100 = g^2$$

- Multiplication of elements:

addition of exponents mod $2^n - 1$ $(2^3 - 1 = 8 - 1) = 7$
 $8 \pmod{7} = 1$

$$\text{e.g. } g^3(g^5) = g^8 = g$$

$$g^2 \cdot g^5 = g^7 = 1$$

$$7 \pmod{7} = 0 \Rightarrow g^0 = 1$$

E.g., consider the elliptic curve $E_{2^m}(a,b) = E_{2^4}(g^4,1)$, specifically

$$y^2 + xy = x^3 + g^4x^2 + 1, \text{ field formed from } x^4 + x + 1$$

$$g^4 = g + 1$$

$$g^1 = g$$

$$g^2 = g^2$$

$$g^3 = g^3$$

$$g^4 = g + 1$$

$$g^5 = g(g^4) = g(g+1) = g^2 + g$$

$$g^6 = g(g^5) = g(g^2 + g) = g^3 + g^2$$

$$g^7 = g(g^6) = g(g^3 + g^2) = g^4 + g^3 = g + 1 + g^3$$

$$g^{14} = g^3 + 1$$

$$g^{15} = g^{14} + g = 1$$

$$g^8 = g(g^7) = g^2 + g + g^4 = g^2 + g + g + 1 = g^2 + 1$$

$$g^9 = g^3 + g$$

$$g^{10} = g^4 + g^2 = g^2 + g + 1$$

$$g^{11} = g^3 + g^2 + g$$

$$g^{12} = g^4 + g^3 + g^2 = g^3 + g^2 + g + 1$$

$$g^{13} = g^4 + g^3 + g^2 + g = g^3 + g^2 + 1$$

$$g^{14} = g^4 + g^3 + g = g^3 + g + 1 + g$$

- Example points on $y^2 + xy = x^3 + g^4x^2 + 1$

E.g. $x=0$ $RS=1$
 $LS=y^2 \therefore y=1$ $(0,1)$ is a point on the curve

$x=g^9$ $RS = (g^9)^3 + g^4(g^9)^2 + 1$
 $= g^{27} + g^{22} + 1$ $2^4-1=15$
 $= g^{12} + g^7 + 1$
 $= \cancel{g^5} + \cancel{g^2} + \cancel{g} + \cancel{1} + \cancel{g^3} + \cancel{g} + \cancel{1} + 1$
 $= g^2 + 1$
 $= g^8$

LHS $\therefore y^2 + g^9y$

$y=g^{10}$ $(g^{10})^2 + g^9(g^{10})$
 $g^{20} + g^{19} = g^5 + g^4 = \cancel{g^2} + \cancel{g} + \cancel{g} + 1$
 $= g^2 + 1 = g^8$

$y=g^{13}$ $(g^{13})^2 + g^9(g^{13}) = g^{26} + g^{22}$
 $= g^{11} + g^7$
 $= \cancel{g^5} + \cancel{g^2} + \cancel{g} + \cancel{g^3} + \cancel{g} + 1$
 $= g^2 + 1 = g^8$

(See Stallings, 6th Ed., Table 10.2, Figure 10.6)